

Part 3: Schema, Semantic Contract & AI Context Quality

Dataset: claims_ledger | Semantic-failure example field: allowed_amount

Semantic-failure example & AI Context Controls: Natalia Walker

A. Schema Expectations done using our main dataset: claims_ledger

Field	Data type	Key/ Nullability	Allowed Values for Each Field
claim_id	String	Primary key, not null	Unique, format C0000000
member_id	String	Foreign key → patient_accounts, not null	Must exist in patient_accounts
provider_id	String	Not null	Format P0000
plan_code	String	Not null	Must match patient_accounts.plan_code
insurance_type	String	Not null	Medicaid, Medicare, Private, Self-Pay
diagnosis_code	String	Not null	Valid ICD-10 code
procedure_code	String	Foreign key → procedure_catalog, not null	Must exist in procedure_catalog
billed_amount	Decimal(10,2)	Not null	≥ 0
allowed_amount	Decimal(10,2)	Not null	≥ 0, ≤ billed_amount
date_of_service	Date	Not null	Valid calendar date, not future-dated
submission_date	Date	Not null	≥ date_of_service
decision_status	String	Not null	Approved, Pending, Denied
patient_state	String	Not null	Valid 2-letter US state code
patient_age	Integer	Nullable	0–120
patient_gender	String	Nullable	Male, Female, Other/Unspecified
provider_specialty	String	Not null	Must match a known specialty list
visit_type	String	Not null	Outpatient, Inpatient, Emergency
length_of_stay	Integer	Nullable	≥ 0 (0 for outpatient/no admission)
chronic_condition_flag	Boolean (0/1)	Not null	0 or 1
is_fraud_flag	Boolean (0/1)	Not null	0 or 1

Schema/version identifier: claims_ledger_schema_v1.0

B. Semantic Contract — claims_ledger

- Business meaning: claims_ledger represents submitted healthcare claims.

Each individual field rows business meaning:

claim_id = unique identifier for one submitted claim; tracks it through its lifecycle

member_id = the covered patient/member the claim was filed for, links to enrollment and coverage terms

provider_id = the provider (physician/facility) who rendered the service and submitted the claim

plan_code = the specific insurance plan governing the claim; determines applicable coverage rules

insurance_type = coverage category (Medicaid, Medicare, Private, Self-Pay) governing which policy rules apply

diagnosis_code = the medical condition being billed for; establishes medical necessity

procedure_code = the specific service performed; determines the applicable fee schedule and coverage rule

billed_amount = the amount the provider originally charged, before any insurer adjustment, in USD

allowed_amount = the amount the payer approved for reimbursement, based on the fee schedule in effect as of date_of_service, in USD

date_of_service = the date the medical service was actually performed; anchors which coverage-policy version applies

submission_date = the date the claim was submitted for adjudication; used to track processing timeliness

decision_status = the payer's coverage decision as of the last system update. The core value the audit decision question evaluates

patient_state = the US state where the patient resides/received care; may affect state-specific coverage rules

patient_age = the patient's age as of claim creation, in whole years (not necessarily their age today)

patient_gender = the patient's recorded gender; relevant to certain clinical/procedure eligibility checks

provider_specialty = the rendering provider's clinical specialty; validates the procedure matches provider expertise

visit_type = the care setting (Outpatient/Inpatient/Emergency) the service was delivered in; affects reimbursement rules

length_of_stay = number of inpatient days, in whole days; relevant to cost/utilization review

chronic_condition_flag = whether the patient has a documented chronic condition; used in risk/cost models

is_fraud_flag = whether the claim has been flagged for suspected fraud; feeds downstream fraud-review workflows

- Units of measure: Monetary fields (billed_amount, allowed_amount) are in US dollars, decimal to the cent. length_of_stay is in whole days. patient_age is in whole years.
- Grain: One row = one claim submitted by a provider on behalf of a member, covering a single procedure/diagnosis pairing on a given date of service.
- Point-in-time / as-of = several fields are only valid relative to a specific date, not as of today. allowed_amount is valid as of date_of_service (the fee schedule in effect that day; decision_status is valid as of the last system update (not necessarily the claim's current real-world status); patient_age is valid as of claim creation (not the patient's current age). The audit decision question: "Was this claim's coverage decision consistent with the policy in effect on the date of service?" -> can only be answered correctly if these as-of dates are checked, not assumed
- Source of truth & freshness: Claims originate from the payer's adjudication system. decision_status is expected to be finalized (non-changing) once approved; all other fields are expected to be stable once submitted.
- Contract/version = claims_ledger_semantic_contract_v1.0; consumers (including AI systems) must not treat any point-in-time field as current-state truth without checking its as-of context (each carries an implicit "valid as of X" condition that isn't stored as a separate timestamp today)